

Health and Economic Returns to Dietary Sodium Reduction

Technical review of the source-split CVD model and preliminary results for prioritized countries

Resolve to Save Lives (RTSL) and University of Washington

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Recap and today's plan

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Recap and today's plan

Recap: where we left off (May 28 call)

From the previous presentation and discussion:

Scope & data

- Fixed to (64) **10 priority countries**.
- Sodium sources: **combined approach** — use country data where available, else the Bhat/Marklund GDP regression (Bhat et al. 2020). Drivers extend beyond GDP (condiments/bouillon in Viet Nam, Nigeria).
- **Do not map interventions to SHAKE** — no longer how the revised SHAKE frames them.

Policies & framing

- **Fiscal policy**: feasibility & generalizability doubts; RTSL interested in salt taxes??
- **LSSS**: focus scenarios 4/5; include potassium.
- **Media**: revised to a UK basis; may be dropped.
- Goal is not only to *maximize* lives saved (e.g. prevention framing for Ethiopia).
- Primary deliverable: **BP reinvestment proposal** (memo, tables + figures), **Sept 2026**.

Today we present the **current build** of the model and **preliminary results** for the prioritized countries.

Today's agenda

1. **Model inputs** — updated sodium source shares by country (now incl. Nigeria/Cameroon) and the applied salt-effect η for each intervention and the full package.
2. **Methods** — the source-split logic (sources $\rightarrow \eta$), the state-transition engine, and the current low-sodium salt substitute (LSS) model.
3. **Preliminary results** — baseline burden, projected trajectories by scenario, and deaths delayed across the 10 prioritized countries.
4. **Limitations.**
5. **Discussion** — pending decisions (fiscal in/out, LSS scenarios, source-share strategy, harmonizing observation periods) and the timeline to the September reinvestment deliverable.

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Model inputs

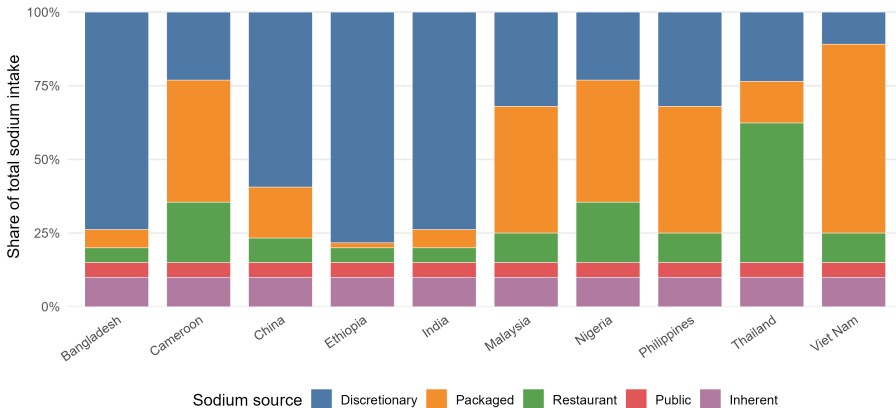
Model inputs at a glance

Input	Role
Baseline transition rates (GBD)	Age $\times$ sex $\times$ year $\times$ cause incidence (IR), case fatality (CF), background & COVID mortality, population (N_x).
BP distribution	Mean/SD systolic BP by age $\times$ sex, raised-BP fraction, diabetes prevalence.
Baseline sodium intake	2024 mean intake, split into five sources.
Sodium source shares	Five-source split of baseline sodium per country (this section).
GBD relative risks	Log-linear RR per 10 mmHg SBP; sets bin-specific baseline incidence.
Ettehad relative risks	RR per 10 mmHg SBP reduction by BP bin $\times$ cause.
Scale-up schedule	Linear ramp 2026 $\rightarrow$ 2030, then held.

The source-split logic is the only change from the earlier single-pool build; everything downstream of η is unchanged.

Sodium source shares by country

Baseline sodium is partitioned into **five mutually exclusive sources** (shares sum to 1). **inherent** sodium is **never targeted**; each policy acts only on the source(s) it can reach.



Country data where available; otherwise predicted from GDP per capita (Bhat et al. 2020). Nigeria (a proxy for Cameroon) use

Applied salt-effect (η) by intervention

Each intervention has a documented **relative reduction** applied to its **target source**.

Intervention	Source	Applied reduction	Note
Public food procurement & service	public	20.0%	
Behaviour change / media campaigns	discretionary	1.0%	
Behaviour change / media campaigns	packaged	1.0%	
Front-of-pack labelling (FOPL)	packaged	13.8%	
Salt targets (reformulation)	packaged	20.0%	
Fiscal policy (salt tax)	packaged	0.0%	
Low-sodium salt substitutes (LSS)	discretionary	15.0%	SSaSS-derived; hypertensive reach; in package

Media acts on discretionary **and** packaged (1.0% each). FOPL, salt targets and fiscal all act on **packaged** and compose multiplicatively when combined: e.g. FOPL (13.8%) + salt targets (20%) $\Rightarrow 1 - (1 - 0.138)(1 - 0.20) = \mathbf{31.0\%}$. **Fiscal is currently 0% — excluded pending the team's decision.**

Illustrative η by intervention and full package

$\text{salteff}(\eta)$ is the **fraction of total intake** each scenario reduces at full implementation — **country-specific**, because it depends on local source shares:

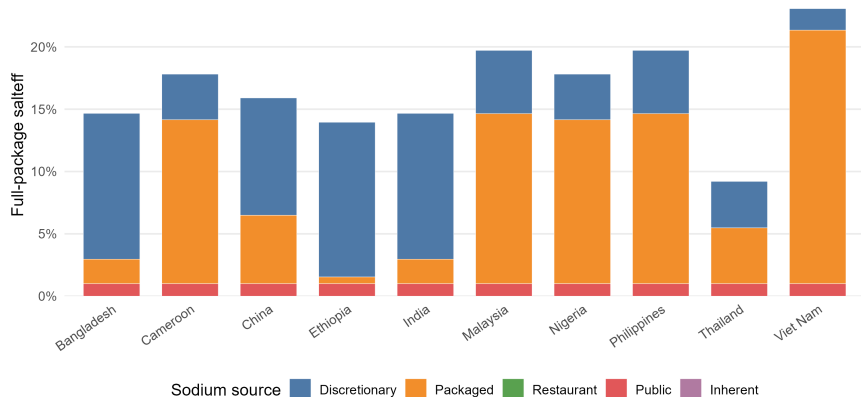
$$\eta = \sum_s \text{share}_s \cdot \left[1 - \prod_i (1 - \text{eff}_{i,s}) \right].$$

Scenario	Viet Nam	Bangladesh	China	Thailand	default
Public food procurement & service	0.0100	0.0100	0.0100	0.0100	0.0100
Behaviour change / media campaigns	0.0075	0.0080	0.0077	0.0038	0.0071
Front-of-pack labelling (FOPL)	0.0885	0.0086	0.0239	0.0195	0.0384
Salt targets (reformulation)	0.1282	0.0124	0.0346	0.0282	0.0557
Fiscal policy (salt tax)	0.0000	0.0000	0.0000	0.0000	0.0000
Low-sodium salt substitutes (LSS)	0.0164	0.1107	0.0891	0.0353	0.0645
Full package	0.2307	0.1466	0.1590	0.0921	0.1665

η is largest where the target source dominates local intake — packaged policies in packaged-heavy Viet Nam; LSS in discretionary-heavy Bangladesh/China. The full-package row composes all active policies.

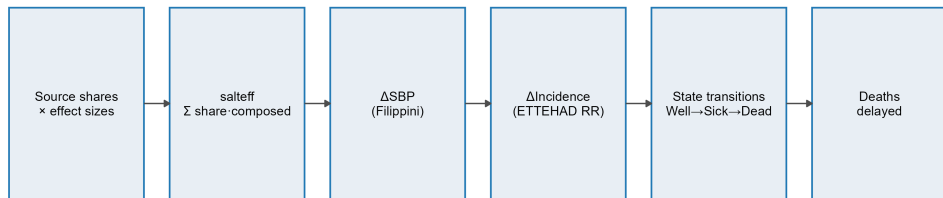
Full-package η : composition by country

Bar height = each country's full-package salteff ; segments show the per-source contributions ($\text{share}_s \times \text{composed}_s$). **No applied effect to restaurants (conservative)**



Methods

The source-split approach: from sources to η



- Split baseline sodium into five sources; apply each policy's reduction **only** to the source(s) it reaches.
- Effects on the **same** source compose multiplicatively — composed_s = $1 - \prod_i (1 - \text{eff}_{i,s})$ — so overlapping packaged policies never double-count or exceed 100%.
- Share-weight and sum to a single total-intake fraction η (salteff), which feeds the unchanged epidemiological pathway.

Worked check (asserted in code): public procurement in Viet Nam touches only the public source (share 0.05, effect 20%)

$$\Rightarrow \eta = 0.20 \times 0.05 = \mathbf{0.0100}.$$

Low-sodium salt substitute (LSS) model — current version

LSS is no longer a placeholder: it is derived from the **Salt Substitute and Stroke Study (SSaSS)** (Neal et al. 2021).

- **Effect size:** a 75% NaCl / 25% KCl substitute delivered an ITT net **-350 mg/day** sodium reduction against a **4.3 g/day** baseline $\Rightarrow$ **15% of discretionary sodium** (sensitivity 0.12–0.18). Applied to the discretionary source.
- **Alternative trial-effect specification:** use the SSaSS outcome RRs directly, applying RR = 0.87 to CVD incidence in hypertensive BP bins (SBP $\geq$ 140) and RR = 0.88 to CFR, scaled by the hypertension-attributable share of CVD deaths.

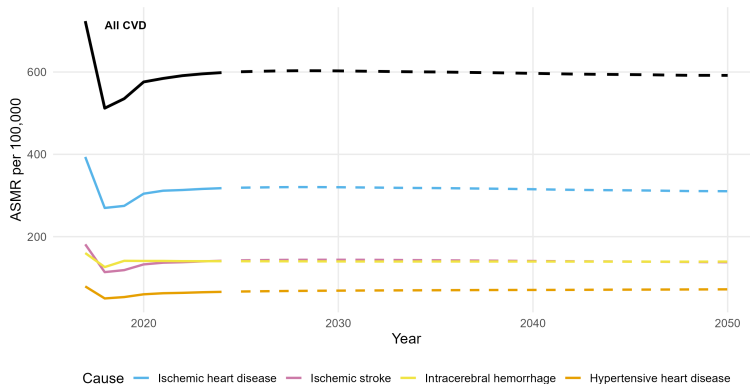
Conservative choices

- **Hypertensive-only reach.** As its own scenario, the reduction applies only to raised-BP bins (SBP $\geq$ 140); in the *full package* it composes across all bins.

Per the May call: include potassium in future work. **Alternative specification (In progress):** direct SSaSS trial-effect pathway, implicitly capturing both sodium reduction and potassium substitution; focus on discretionary scenarios 4/5 (Neal et al. 2021; Huang et al. 2026).

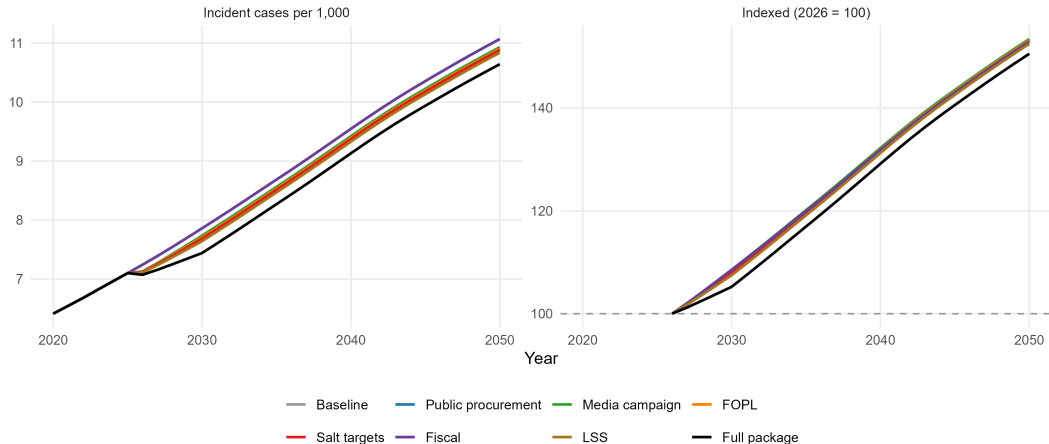
Baseline and projections

Baseline CVD burden, 2017–2050



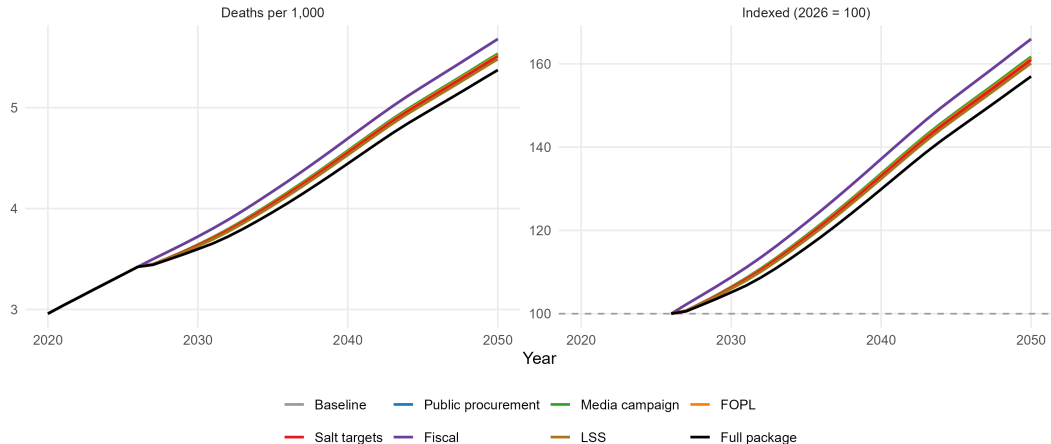
Age-standardised mortality (WHO standard, ages 20–95+) declines with forecasted background-mortality and case-fatality trends; absolute CVD deaths still rise with population growth and ageing. Solid = historical, dashed = projected.

Projected incidence by scenario



Incident CVD cases per 1,000 (left) and indexed to 2026 = 100 (right). Scenarios coincide until scale-up begins, then fan out; the **full package** falls fastest. **Fiscal overlaps baseline** because its applied η is currently 0%.

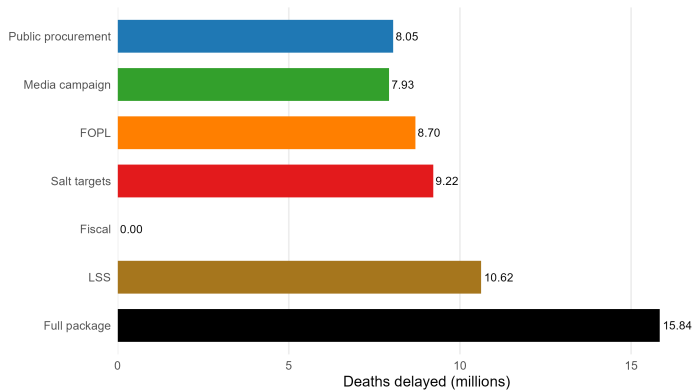
Projected deaths by scenario



CVD deaths per 1,000 (left) and indexed to 2026 = 100 (right). Because mortality responds to the **prior** year's incidence, the scenarios' divergence lags the incidence divergence by about a year.

Main results

Deaths delayed by scenario, 2026–2050



Headline

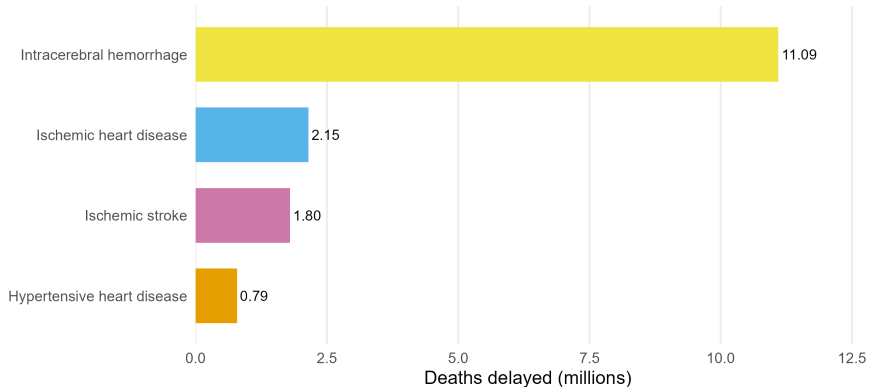
The **full package** delays 15.84 million CVD deaths across 10 countries. 4.6

Deaths delayed by scenario — detail

Scenario	Deaths delayed (millions)	Avg per year (thousands)	% of baseline
Public food procurement & service	8.051	322.0	2.32%
Behaviour change / media campaigns	7.931	317.3	2.28%
Front-of-pack labelling (FOPL)	8.696	347.8	2.50%
Salt targets (reformulation)	9.221	368.8	2.65%
Fiscal policy (salt tax)	0.000	0.0	0.00%
Low-sodium salt substitutes (LSS)	10.618	424.7	3.05%
Full package	15.835	633.4	4.55%

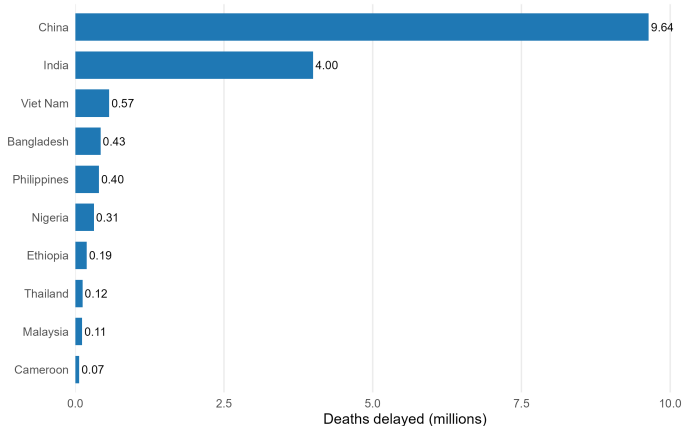
Among single policies, **LSS** delays the most (10.62 M) — it targets discretionary salt, which dominates intake in the most populous countries — followed by **salt targets** (9.22 M), **FOPL** (8.70 M), **public procurement** (8.05 M) and **media** (7.93 M). **Fiscal** delays 0.00 M (applied $\eta = 0\%$).

Full package: deaths delayed by cause



Ischemic heart disease and stroke dominate the delayed deaths, tracking their share of the baseline CVD burden and their SBP-mediated relative risks.

Full package: deaths delayed by country



Absolute counts are dominated by the most populous countries (China, India); per-capita impact depends on baseline burden and local source structure.

Limitations

Limitations and interpretation

- **Deaths delayed, not averted.** Case fatality is held at baseline, so the intervention lowers incidence and shifts mortality later; totals depend on the accounting window.
- **Point estimates only.** The per-source reductions are deterministic — no probabilistic sensitivity analysis yet.
- **LSS is a conservative lower bound** — sodium arm only, no potassium channel; hypertensive-only reach as an individual scenario.
- **Fiscal policy currently excluded** pending the team's decision. (applied $\eta = 0$)
- **Provisional source shares.** Nigeria/Cameroon values are working estimates; several others rest on single-study or older survey data.
- **Heterogeneous observation periods.** Documented effects span 1–20 year windows and are not yet harmonized onto a common footing.
- **Independent-cause structure.** The four causes share one population but are modelled independently.

Discussion and next steps

Pending decisions — for discussion

Open items

1. **Fiscal policy** — in or out? Currently $\eta = 0$
2. **LSS scenarios** — confirm discretionary scenarios 4/5 (prevalent or treated??); add a potassium channel to capture the full SSaSS effect?
3. **Observation-period harmonization** — effects measured over 1–20 yr. Apply as end-state at full scale-up, linearly over the observed window, or via a country-tier heuristic?
4. **Source-of-sodium strategy** — how hard to push for country-specific shares vs. the Bhat/Marklund GDP regression (Bhat et al. 2020)?.
5. **Policy framing** — lives-saved maximization vs. a prevention/counterfactual framing (e.g. Ethiopia).
6. **Media / behavior change** — keep or drop?

Next steps and timeline

- **Lock the inputs** — finalize source shares (incl. Nigeria/Cameroon), confirm effect sizes, and resolve the fiscal decision.
- **Add the economic layer** — policy costing plus benefit–cost outputs (NPV, BCR) and DALYs.
- **Primary deliverable** — the **BP reinvestment proposal** memo (tables + figures), target **September 2026**.
- **Paper** is secondary; coordinate with Matti on the Bhat/Marklund inputs to avoid scooping the forthcoming paper.

Discussion

Where should we push hardest before the reinvestment deliverable — e.g data (source shares)?

Appendix

Appendix: sodium-reduction levers are source-specific

Institutional framing

- WHO/PAHO frame sodium reduction as a **government-led policy package**, not individual education alone.
- Core target: shift the **food environment** so lower-sodium choices become the default.
- Stronger instruments are increasingly favored: **mandatory standards, procurement rules, warning labels, reformulation targets, and monitoring/enforcement.**

How this maps to the model

- Each policy acts only on the sodium source it can plausibly reach:
 - procurement → public meals
 - FOPL / salt targets / fiscal → packaged foods
 - LSS → discretionary salt
 - media → discretionary + packaged
- Effects are composed by source, then translated into total sodium reduction η .

Interpretation

The full package is not a single intervention: it represents a coordinated regulatory, market, and behavior-change strategy that reduces sodium through the sources that dominate intake in each country.

Sources: WHO sodium-reduction guidance and SHAKE package; WHO public procurement framework; WHO global sodium benchmarks; PAHO front-of-package labelling and nutrient profile model; CDC Food Service Guidelines.

Appendix: per-intervention methods

The following slides document each intervention's **evidence base** and the **applied** η used in the model. Applied values are produced by `reports/report.RMD` and `mirror code/07_run_interventions.R`; study effect sizes are cited from source.

Also in this appendix: documented source-share values (Table M1) and the per-scenario η decomposition by country.

A. Public food procurement

Target source: public (institutional meals). **Applied η :** 20% of the public source.

- **Evidence:** Gase et al. (2011) — modeling study, LA County. Weighted **-233 mg/day** ($\approx 7\%$ of public-meal sodium).
- **Source adjustment:** for a country with share x of sodium from institutional meals, population impact $= 0.20 \times x$. Worked: Viet Nam $= 0.20 \times 0.05 = 0.0100$.
- **Caveats:** modeled (not observed); US-specific setting; small affected population share.

B. Behavior change / mass media

Target source: discretionary + packaged. **Applied η : 1.0% to each.**

- **Evidence:** Do et al. (2016) — COMBI Vietnam, **5%** fall in mean sodium after a 1-year multi-component campaign. Collins et al. (2014) — UK economic evaluation, **2%** [1–5%].
- The estimate was revised toward a UK basis; a conservative 1.0% is applied to each of discretionary and packaged.

Open question

RTSL funds **policy change, not mass media for behavior change** (Laura). Keep this lever in the model, or drop it?

C. Low-sodium salt substitutes (LSS)

Target source: discretionary salt. **Applied η : 15% of discretionary** (sensitivity 0.12–0.18).

- **Evidence:** Neal et al. (2021) (SSaSS) — -350 mg/day ITT sodium reduction vs. a 4.3 g/day baseline; observed -3.34 mmHg SBP (much of it potassium-driven, +803 mg/day).
- **Modelling:** sodium arm only (no K^+ channel) $\Rightarrow$ conservative lower bound; hypertensive-only reach as an individual scenario.
- **Scenarios** (Huang et al. 2026): focus on discretionary-only 2 (all), 4 (diagnosed HTN), 5 (treated HTN).

D. Front-of-pack labeling

Target source: packaged (estimate already from packaged-food purchases). **Applied η :** 13.8% of packaged.

- **Evidence:** Taillie et al. (2024) — Chile's mandatory warning-label law (observed). **-73.3 mg/capita/day (-13.8%)** in sodium purchases (2,844 households, 2013–2019).
- **Caveat:** purchases-to-intake translation is uncertain.
- **Alternative:** Mexico found $\approx -7\%$ in packaged sales (≈ -38 mg/capita/day).

E. Mandatory salt targets (reformulation)

Target source: packaged. **Applied η :** **20% of packaged** (conservative; 15% floor for sensitivity).

- **Policy:** South African mandatory maximum sodium limits (2013; targets implemented 2016 and 2019).
- **Evidence:** Strauss-Kruger et al. (2022) — African-PREDICT, **−15.2%** sodium; baseline/follow-up straddle the target dates, so likely an **underestimate**.
- **Alternatives:** (Gaziano et al. 2025) −8.4% (rural, spot urine); (Charlton et al. 2021) −18.4%; UK (He, Pombo-Rodrigues, and MacGregor 2013; Song et al. 2023); Mexico modeling (Muciño-Sandoval, Barrientos-Gutiérrez, et al. 2025).

F. Fiscal policy / taxation

Target source: packaged. **Applied η : 0% — currently excluded (see below).**

- **Evidence:** Saxena et al. (2025) — Philippines, modeled 20% tax on foods high in salt/sugar; **−100 mg/day (−10%)** at year 20.
- **Why excluded (for now):** political feasibility of food-category taxes; generalizability of a country-specific modeled estimate; RTSL not interested in salt taxes (May call).
- **Alternatives:** India (Roche et al. 2026); Australia (Davies et al. 2026; Cobiac et al. 2017); systematic review (Dodd et al. 2020) (1–11% reduction).

Appendix: sodium source shares — documented values

Country	Discretionary	Packaged	Restaurant	Public	Inherent
Viet Nam	0.109	0.641	0.100	0.050	0.100
Philippines	0.320	0.430	0.100	0.050	0.100
Bangladesh	0.738	0.062	0.050	0.050	0.100
China	0.594	0.173	0.083	0.050	0.100
Ethiopia	0.783	0.017	0.050	0.050	0.100
India	0.738	0.062	0.050	0.050	0.100
Malaysia	0.320	0.430	0.100	0.050	0.100
Thailand	0.235	0.141	0.473	0.050	0.100
Nigeria	0.231	0.415	0.205	0.050	0.100
Cameroon	0.231	0.415	0.205	0.050	0.100
default	0.430	0.279	0.142	0.050	0.100

Fractions of total intake, renormalised to 1.0. The default row (mean of the ten) applies to any country without its own row.

Appendix: η decomposition by scenario and country



Per-scenario salteff for every country, stacked by source contribution (note free y-axes). Fiscal is empty because its applied

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